CHAPTER 2 · GRADE 4 · WEEK 1

Moving Objects Have Energy

Motion · How do we know a moving object has energy?



Unit 1 · Energy Transfer & Transformation

TURN & TALK BQ

Big Question

How are energy, change, and movement of objects related?

Keep it in mind all chapter. We will come back to it.

Think: What is ONE change a moving object can cause?

WRITE N

BQ

Big Question – My first thinking

Guided Notes p. 1

How are energy, change, and movement of objects related?

My first thinking

Write 1–2 quick ideas. No wrong answer yet.

Marble Collision — before

One marble sits still. Another marble rolls toward it.

BEFORE

A target marble is still.

A moving marble rolls toward it.

Observe closely. What do you see BEFORE the marbles touch?

PREDICT D

Predict: what will happen?

Think before the moving marble reaches the still marble.

When the moving marble hits the still marble, the still marble will...

get _____

Make a prediction. Say it to your partner.

What changed?

BEFORE

target marble still

AFTER

target marble now moving

CHANGES

- the still marble starts to move

How do we know the marble had energy?

Energy can be invisible. We look for the change.



The still marble started to move.
That change is evidence that the moving marble had energy.

V

Vocabulary – Motion

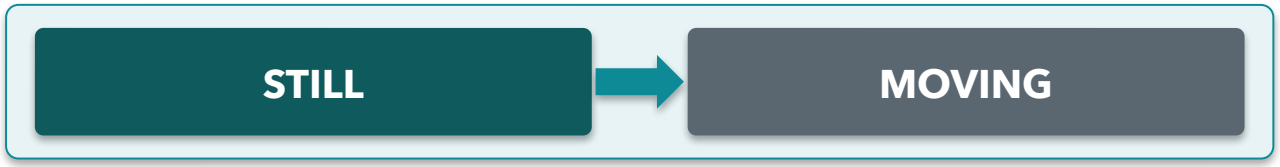
Reader p. 7

VOCABULARY

TEACH

motion

the process of an object changing position



Reader p. 7

WRITE N

V1

Vocabulary – Motion

Guided Notes · Reader p. 7

motion – the process of an object changing _____ .

Write the missing word in the blank.

CHECK N

V1

Vocabulary – Motion

Guided Notes · Reader p. 7

motion – the process of an object changing **position** .

position

What can a moving object do?

Reader p. 8 — think before you write.

Some objects start their own motion.
A ball does not.

Players cause their own motion. A ball does not. Something else puts the ball into motion.

N1 Write the Missing Words

Guided Notes · Reader p. 8

A moving object can cause a _____ .

Players cause their own motion. A ball does not. Something else puts the ball into motion.

Some objects cause their own motion. Other objects are put into motion by _____ motion.

N1 Moving Objects Can Cause Changes

Guided Notes · Reader p. 8

A moving object can cause a **change** .

Players cause their own motion. A ball does not. Something else puts the ball into motion.

Some objects cause their own motion. Other objects are put into motion by **outside** motion.

change · outside

V

Vocabulary — Energy of Motion

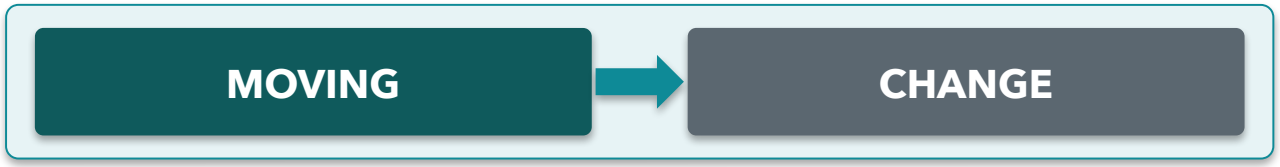
Reader p. 9

VOCABULARY

TEACH

energy of motion

the energy an object possesses while it is moving



Reader p. 9

WRITE N

V2

Vocabulary – Energy of Motion

Guided Notes · Reader p. 9

energy of motion – the energy an object possesses while it is _____ .

Write the missing word in the blank.

CHECK N

V2

Vocabulary — Energy of Motion

Guided Notes · Reader p. 9

energy of motion — the energy an object possesses while it is **moving** .

moving

Moving objects have energy

Reader p. 9 – energy is the ability to cause change.



Energy is the ability to cause change.

A moving object causes change, so it has energy.

N2

Moving Objects Have Energy

Guided Notes · Reader p. 9

All moving objects have energy of _____ .

Energy is the ability to cause change. A moving object causes change, so it has energy.



N2

Marble Collision — Observe

Guided Notes · Reader p. 9

OBSERVE — one moving marble rolls into a target marble.

BEFORE	AFTER
draw	draw

What changed?

N2

Moving Objects Have Energy

Guided Notes · Reader p. 9

All moving objects have energy of **motion** .

Energy is the ability to cause change. A moving object causes change, so it has energy.



motion
"The target marble changed from sitting still to moving."

R

People use the energy of motion

Reader pp. 10–12 — moving objects do useful work.

hammer (swinging)

pendulum (swinging)

wind (moving air)

A swinging hammer, a swinging pendulum, and moving air (wind) all use the energy of motion to cause a change.

N3

People Use the Energy of Motion

Guided Notes · Reader pp. 10–12

Write or draw the change each moving object causes.

MOVING OBJECT	WHAT CHANGE DOES IT CAUSE?
hammer (swinging)	
pendulum (swinging)	
wind (moving air)	

N3

People Use the Energy of Motion

Guided Notes · Reader pp. 10–12

MOVING OBJECT	WHAT CHANGE DOES IT CAUSE?
hammer (swinging)	drives a nail into the wood
pendulum (swinging)	turns the clock gears
wind (moving air)	spins the turbine blades

WRITE N

CHECK

Check Your Thinking

Guided Notes · Reader pp. 7–12

How do we know a moving object has energy?

Start with: "We know because the moving object causes …"

We know because it causes a change

SAMPLE

"We know because the moving object causes a change – like a moving marble making another marble move."

Any change a moving object causes is evidence that it has energy.

EVIDENCE C

Which change shows the energy of motion?

A moving marble makes a still marble move.

A rock keeps sitting still on the ground.

A pencil rests on a desk.

Look for a change – that is the evidence of the energy of motion.

WRITE N

BQ

BQ · Now I Think

Guided Notes p. 3

BIG QUESTION

How are energy, change, and movement of objects related?

Now I think...

C

Review 1 – concept

A moving object has energy. What is this energy called?

A stored energy

B energy of motion

C energy of light

D energy of sound

CHECK C

Review 1 – answer

A moving object has energy. What is this energy called?

A stored energy

B energy of motion

C energy of light

D energy of sound

Energy of motion (Reader p. 9).

C

Review 2 – visual

A marble rolls and hits a still marble. What change do you observe?

A Both marbles stop moving

B The still marble starts to move

C The marbles change color

D Nothing happens

CHECK C

Review 2 – answer

A marble rolls and hits a still marble. What change do you observe?

A Both marbles stop moving

B The still marble starts to move

C The marbles change color

D Nothing happens

The still marble starts to move (Reader p. 8).

C

Review 3 — scenario

A swinging hammer hits a nail. What change does the hammer cause?

A The hammer turns to liquid

B The hammer floats away

C The nail is driven into the wood

D The nail melts

CHECK C

Review 3 – answer

A swinging hammer hits a nail. What change does the hammer cause?

A The hammer turns to liquid

B The hammer floats away

C **The nail is driven into the wood**

D The nail melts

The hammer drives the nail in (Reader p. 10).

C

Review 4 — application

How do we know a moving object has energy if we cannot see the energy itself?

A We can always see energy directly

B We guess without any evidence

C Moving objects never have energy

D We observe the change it causes

CHECK C

Review 4 — answer

How do we know a moving object has energy if we cannot see the energy itself?

A We can always see energy directly

B We guess without any evidence

C Moving objects never have energy

D We observe the change it causes

We know because it causes a change.

C

Review 5 — scenario

Wind (moving air) spins windmill blades. What does this show?

A Moving air has energy of motion

B Still air is stronger

C Windmills work without energy

D Blades move on their own

CHECK C

Review 5 – answer

Wind (moving air) spins windmill blades. What does this show?

A **Moving air has energy of motion**

B Still air is stronger

C Windmills work without energy

D Blades move on their own

Moving air has energy of motion (Reader p. 12).